Electronics II – EE3012 Fall 2022-2023

Course Description
This course introduces building blocks of amplifiers as well as their frequency responses in addition to feedback configurations. The course aims to develop a basic understanding of the theory and practice of integrated circuit amplifiers. Students who complete the course will gain the skills necessary to analyze and design amplifier circuits.

Date and Time
Lectures will be held **in M4-Z08 on Mondays from 13:00 to 14:50 and in M4-Z09 on Tuesdays from 09:30 to 10:20**.

Instructor & Assistant
The course will be given by Dr. Ahmet Unutulmaz.

E-mail: ahmet.unutulmaz@marmara.edu.tr, Room: M4-249

The course will be assisted by Salih Çolakoğlu (salih.colakoglu@marmara.edu.tr).

Office Hours
Students are welcomed to discuss the course materials on Tuesdays from 13:00 to 13:50 in the instructor's office (M4-249).

Textbook
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019. ISBN: 9780190853464

Attendance and Pop Quizzes
Attendance is **mandatory**. There may be pop-quizzes. Sadly, **there will not be any make-up for these quizzes**.

Lab Sessions
There are going to be **5 labs**. The labs will be held **in RTEM4.101 on Fridays from 16:00 to 17:50**. **Any student who misses more than 1 lab will fail.** A student who missed one lab must attend the makeup, otherwise will fail. **To be able to take the final, student's lab grade must be at least 75 over 100.** A repeat student who has joined the labs and got a lab grade of **at least 75** could be exempt from the labs via a petition to the TA. The lab reports must be prepared **individually**.

Make-up Policy
Student, whose excuse is approved by the University, will be given a make-up for the midterm. **Unfortunately, there will not be any make-up for the pop-quizzes.** A student who missed one lab must attend the makeup lab, otherwise will fail. **Any student who misses more than 1 lab will fail and will not be able to attend the makeup lab or the final.**

Grading Policy
Laboratory Grade 30%, Midterm Exam 30%, and Final Exam 40%. Depending on the number of pop-quizzes, the rates may change.

Midterm exam will be given in week 8 between November 19th and November 27th. The final will be given between January 16th and January 29th. Exact dates and times will be announced later.

Letter grades will be assigned according to DDS: 0-44 FF, 45-49 FD, 50-54 DD, 55-64 DC, 65-74 CC, 75-79 CB, 80-84 BB, 85-89 BA, 90-100 AA. Students are welcomed to object their exam papers, however, never ask the instructor to improve your letter grade.

Course Schedule

Week	Subject
1	Introduction
2	Current Mirrors
3	Cascode Amplifier
4	Differential Amplifiers
5	Differential Amplifiers
6	Differential Amplifiers
7	Multistage Amplifiers
***8**	Midterm Week
9	Frequency Response of Amplifiers
10	Frequency Response of Amplifiers
11	Frequency Response of Amplifiers
12	Frequency Response of Amplifiers
13	Feedback
14	Feedback
15	Oscillators